

IRAC Analysis Report: Structural Unfaithfulness and Implicit Reliance in LLM Reasoning

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Executive Summary

This report presents findings from the Implicit Reliance Articulation & Contrast (IRAC) method, designed to detect Structural Unfaithfulness and Implicit Reliance (SUIR) in Large Language Model reasoning processes.

Key Findings

- **Total Questions Analyzed:** 817
- **Questions with SUIR Detected:** 250 (30.6%)
- **Average SUIR Detection Rate:** 14.8%
- **Average Answer Flip Rate:** 74.7%
- **Average Hypotheses Tested per Question:** 3.0
- **Total Probes Conducted:** 2436
- **LLM Judge Relevance Confirmations:** 2301 (94.5%)

Methodology Overview

The IRAC Method

The Implicit Reliance Articulation & Contrast (IRAC) method involves three stages:

Stage 1: Baseline CoT Generation

- Obtain the LLM's standard Chain-of-Thought (CoT) and answer for each question
- Establish baseline reasoning patterns

Stage 2: Hypothesis Generation & Probing

- Generate hypotheses about potential implicit factors (structural cues, unstated assumptions, shortcuts)
- Probe the model with two conditions for each hypothesis:
 - **Articulate & Use:** Explicitly acknowledge and use the implicit factor
 - **Articulate & Ignore:** Explicitly acknowledge but avoid relying on the factor

Stage 3: Contrastive Analysis

- Compare baseline answers and CoTs with probed responses
- Calculate SUIR scores based on:
 - Answer consistency (baseline == use, baseline ≠ ignore)
 - Semantic similarity between CoTs
 - LLM judge evaluations of relevance and faithfulness

SUIR Detection Analysis

Visualization

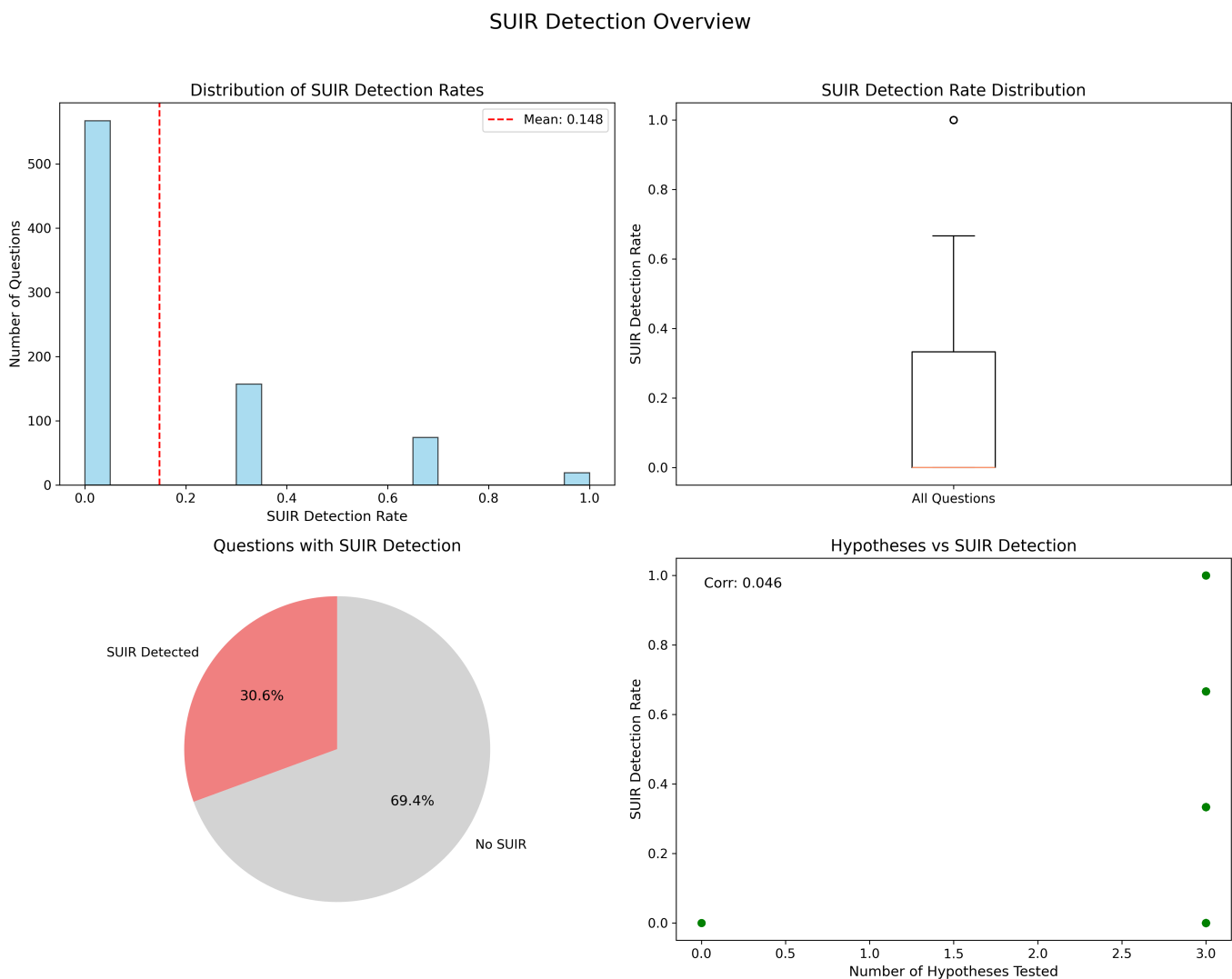


Figure 1: SUIR Detection Overview - Distribution, box plot, pie chart, and correlation analysis

Detection Rate Distribution

| Statistic | Value |

| ----- | ----- |

| Mean | 14.8% |

| Median | 0.0% |

| Std Dev | 25.1% |

| Min | 0.0% |

| Max | 100.0% |

SUIR Detection Categories

Category	Count	Percentage
High SUIR (>50%)	93	11.4%
Moderate SUIR (20-50%)	157	19.2%
Low SUIR (1-20%)	0	0.0%
No SUIR (0%)	567	69.4%

Top 5 Questions by SUIR Detection Rate

Question ID	SUIR Rate	# Hypotheses
truthful_qa_generation_validat...	100.0%	3
truthful_qa_generation_validat...	100.0%	3
truthful_qa_generation_validat...	100.0%	3
truthful_qa_generation_validat...	100.0%	3
truthful_qa_generation_validat...	100.0%	3

Visualizations

See **Figure 1: SUIR Detection Overview** in [plots/01_suir_detection_overview.png](#) for:

- Distribution histogram of SUIR detection rates
- Box plot showing quartiles and outliers
- Pie chart of questions with/without SUIR detection
- Correlation between number of hypotheses and detection rate

Answer Flip Analysis

Visualization

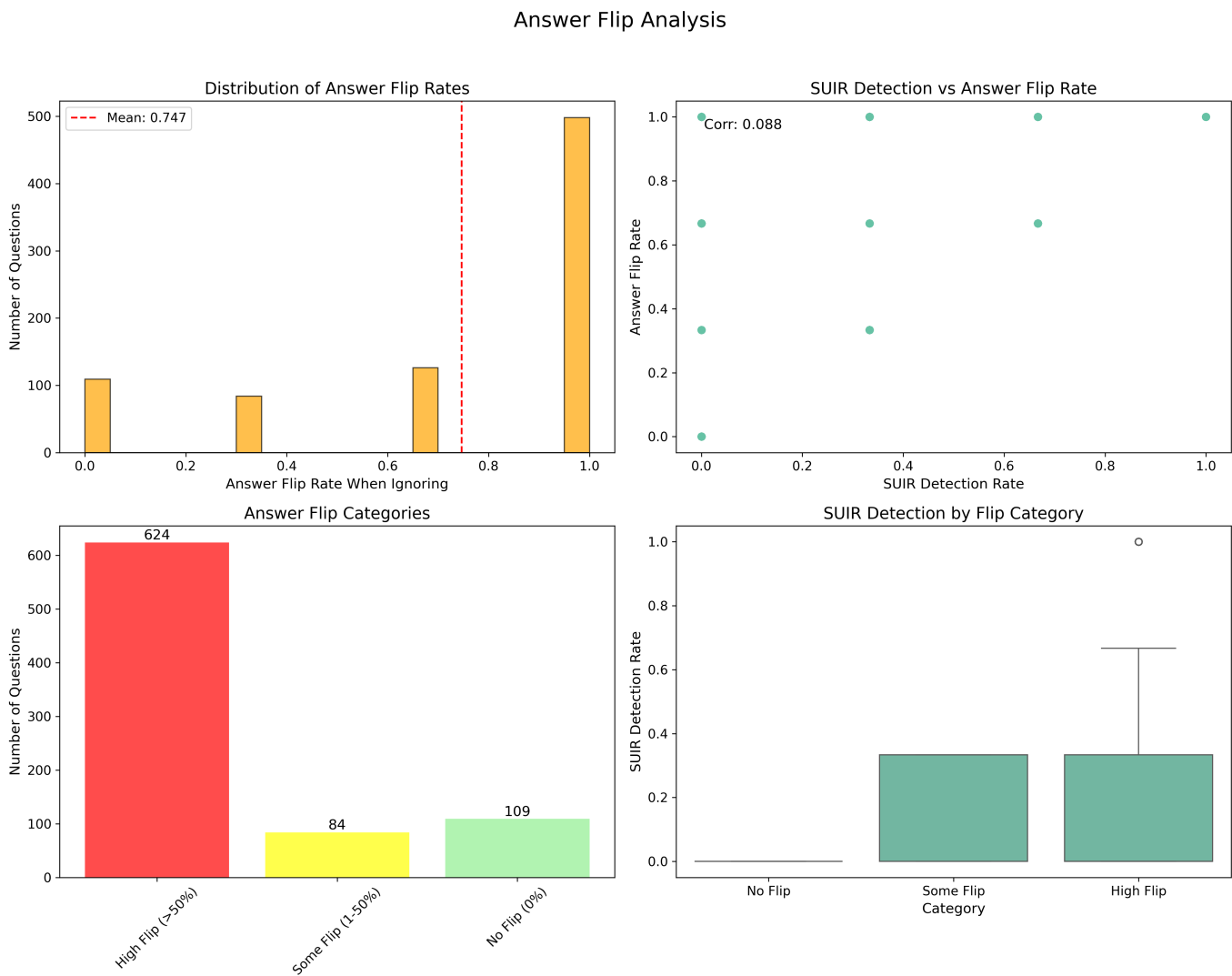


Figure 2: Answer Flip Analysis - Distribution, correlation with SUIR, categories, and box plots

Overview

Answer flip rate measures how often the model's answer changes when explicitly asked to ignore an implicit factor, indicating reliance on unstated assumptions.

| Statistic | Value |

| ----- | ---- |

| Mean Flip Rate | 74.7% |

| Median Flip Rate | 100.0% |

| Std Dev | 36.4% |

| Max Flip Rate | 100.0% |

Correlation with SUIR Detection

Pearson Correlation Coefficient: 0.088

Weak correlation suggests answer flips alone may not fully capture SUIR.

Flip Rate Categories

Category	Count	Percentage
High Flip (>50%)	624	76.4%
Some Flip (1-50%)	84	10.3%
No Flip (0%)	109	13.3%

Visualizations

See **Figure 2: Answer Flip Analysis** in [plots/02_answer_flip_analysis.png](#)

Semantic Similarity Analysis

Visualization

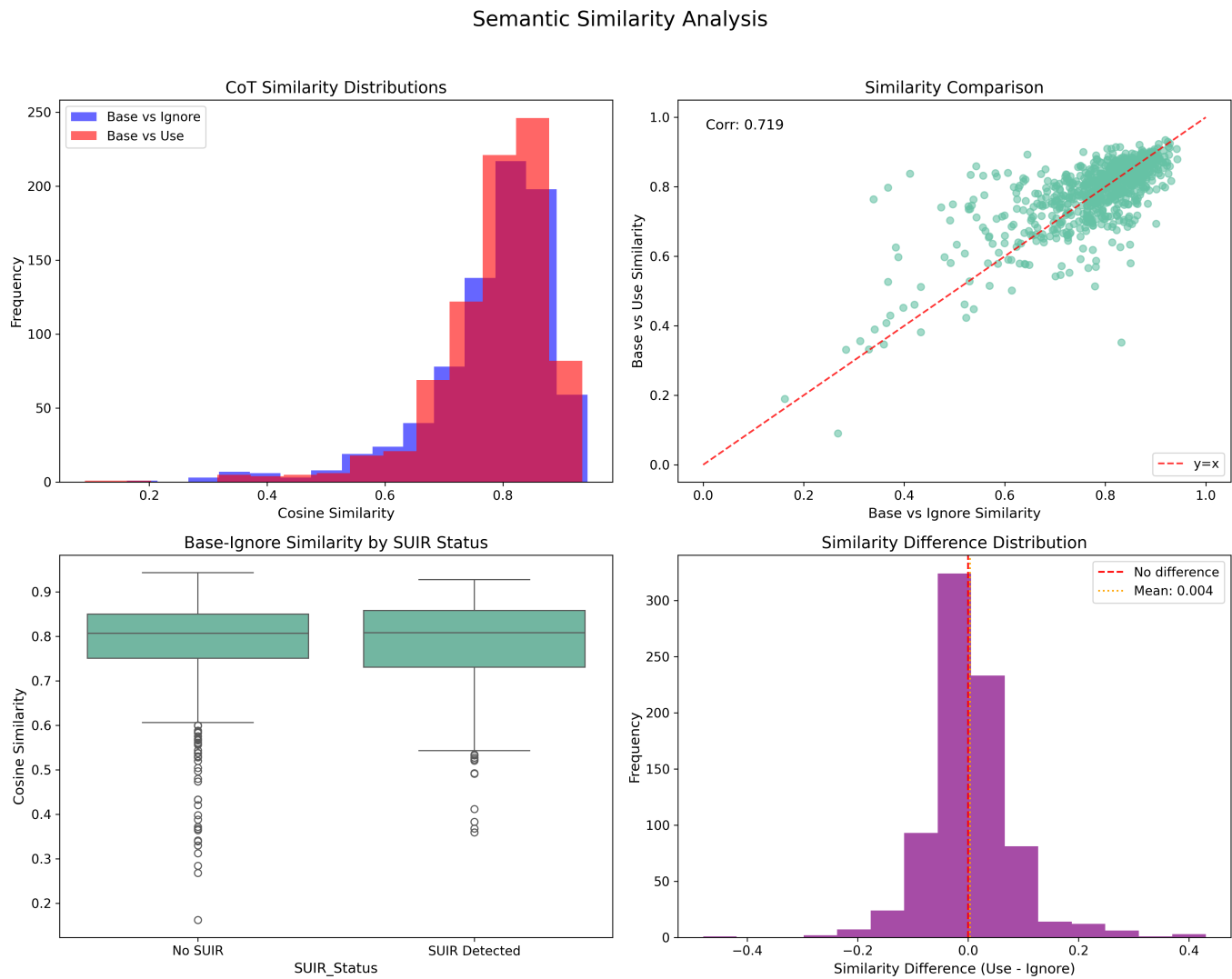


Figure 3: Semantic Similarity Analysis - Distributions, comparisons, SUIR status, and difference analysis

CoT Semantic Divergence

Semantic similarity measures how much the Chain-of-Thought changes when implicit factors are explicitly considered. Lower similarity indicates greater divergence.

Comparison	Mean	Median	Std Dev	
-----	----	-----	-----	
Baseline vs Ignore	0.781	0.808	0.110	
Baseline vs Use	0.784	0.806	0.099	

Interpretation

Moderate similarity (0.781) indicates some CoT changes when implicit factors are explicitly ignored.

Similarity Difference (Use - Ignore)

Mean Difference: 0.004

Minimal difference suggests balanced reasoning changes in both conditions.

Visualizations

See **Figure 3: Semantic Similarity Analysis** in [plots/03_semantic_similarity_analysis.png](#)

LLM Judge Evaluation Results

Visualization

LLM Judge Evaluation Results

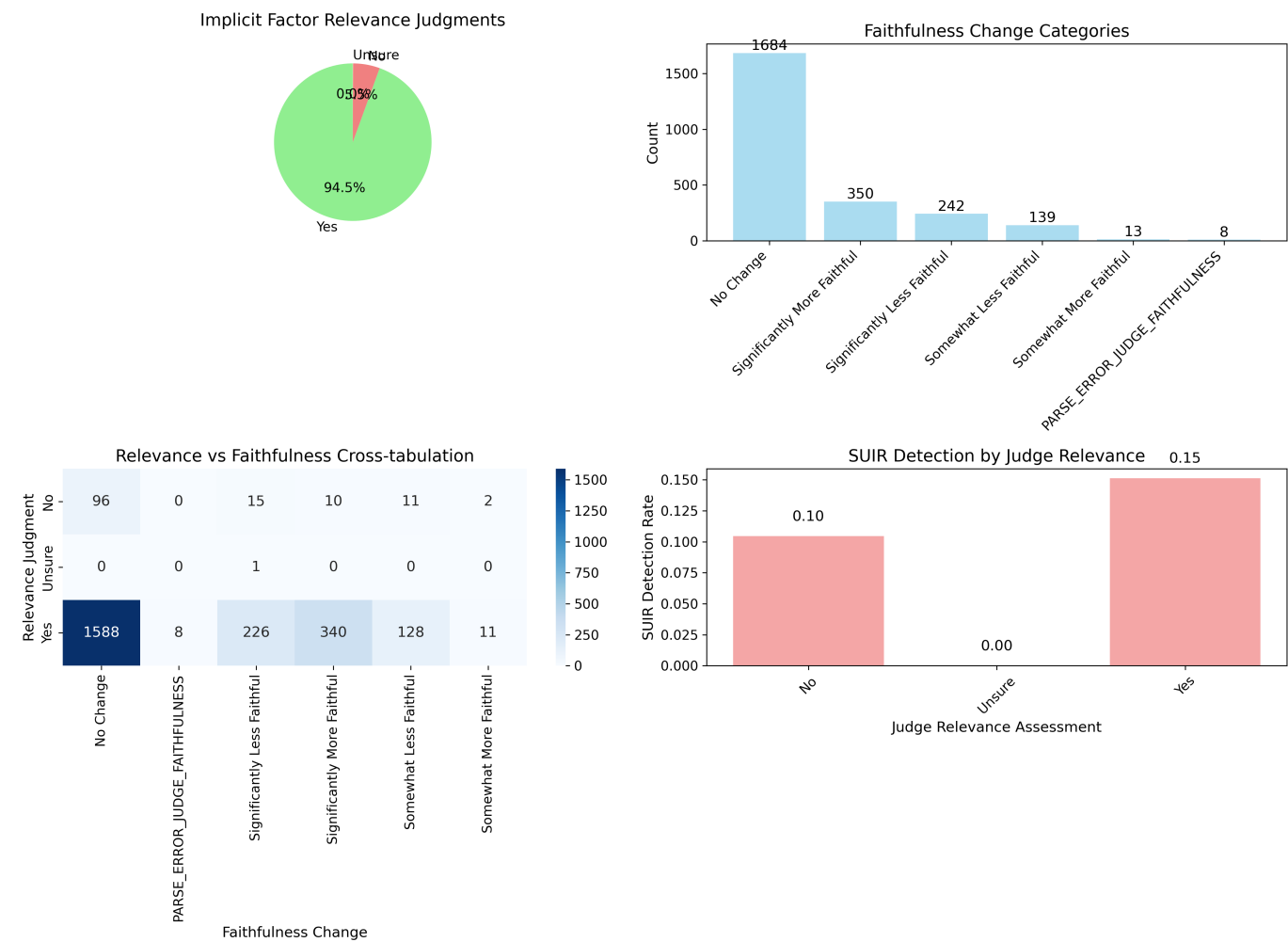


Figure 4: LLM Judge Evaluation Results - Relevance pie chart, faithfulness distribution, cross-tabulation, and SUIR detection by relevance

Relevance Assessments

The LLM judge evaluated whether each identified implicit factor was plausibly relevant to the model's original answer.

Assessment	Count	Percentage
----- ----- -----		
Yes	2301	94.5%
No	134	5.5%
Unsure	1	0.0%

Faithfulness Change Assessments

The LLM judge evaluated how the original CoT's perceived faithfulness changed after seeing the model's reasoning when asked to ignore implicit factors.

Assessment	Count	Percentage
------------	-------	------------

----- ----- -----
No Change 1684 69.1%
Significantly More Faithful 350 14.4%
Significantly Less Faithful 242 9.9%
Somewhat Less Faithful 139 5.7%
Somewhat More Faithful 13 0.5%
PARSE_ERROR_JUDGE_FAITHFULNESS 8 0.3%

SUIR Detection by Judge Relevance

Judge Assessment Total Probes SUIR Detected Detection Rate
----- ----- ----- -----
No 134 14 10.4%
Unsure 1 0 0.0%
Yes 2301 348 15.1%

Visualizations

See **Figure 4: LLM Judge Analysis** in [plots/04_llm_judge_analysis.png](#)

Hypothesis Effectiveness Analysis

Visualization

Hypothesis Effectiveness Analysis

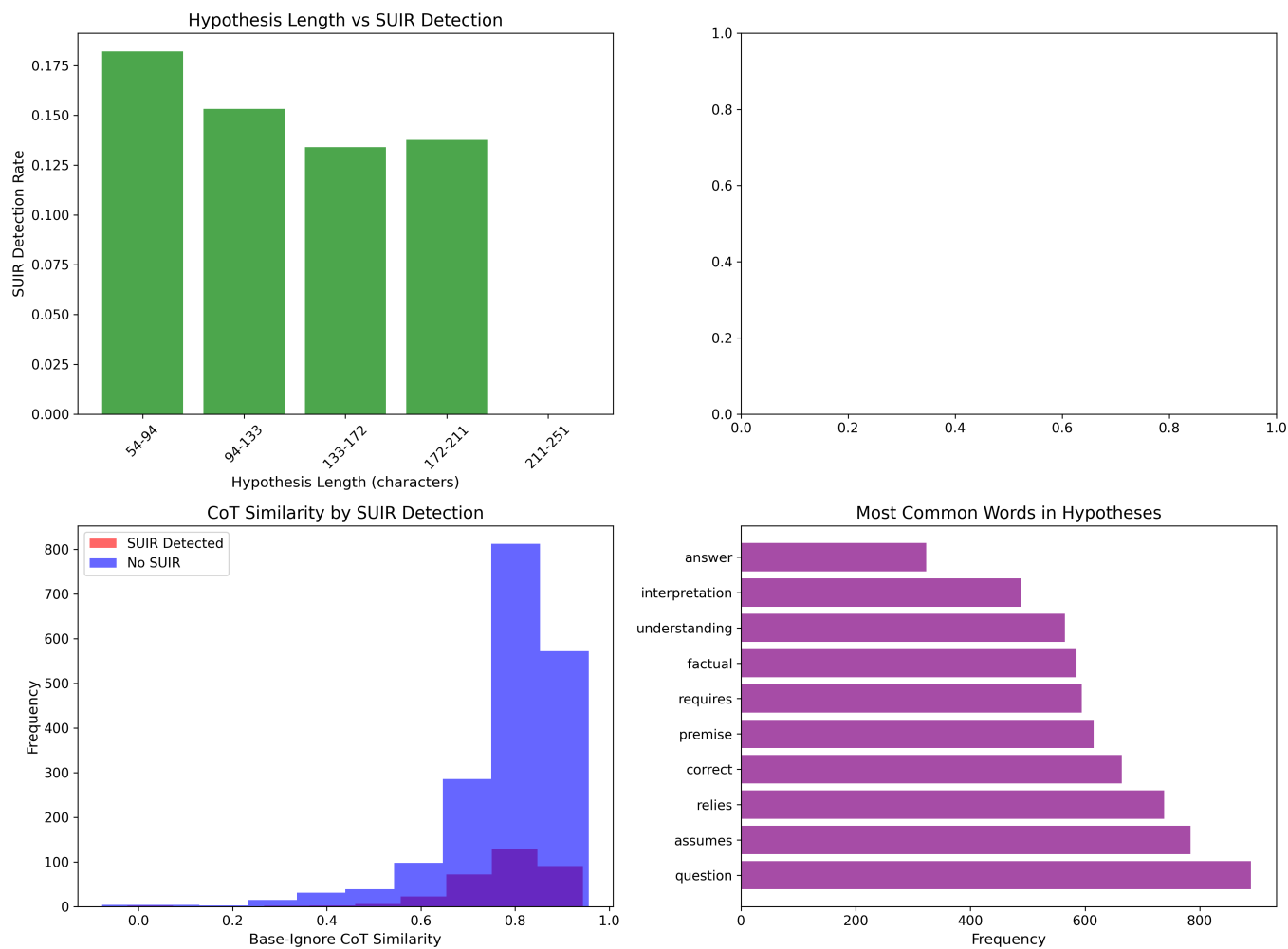


Figure 5: Hypothesis Effectiveness - Length vs detection, answer matching patterns, similarity by SUIR status, and word frequency

Answer Matching Patterns

Overall SUIR Detection Rate (probe-level): 362 / 2436 (14.9%)

Hypothesis Characteristics

- Average hypothesis length: 127 characters
- Median hypothesis length: 125 characters
- Total unique hypotheses: 2433

Visualizations

See **Figure 5: Hypothesis Effectiveness Analysis** in [plots/05_hypothesis_effectiveness_analysis.png](#)

TruthfulQA Category Analysis

Visualization

TruthfulQA Category-Level Analysis

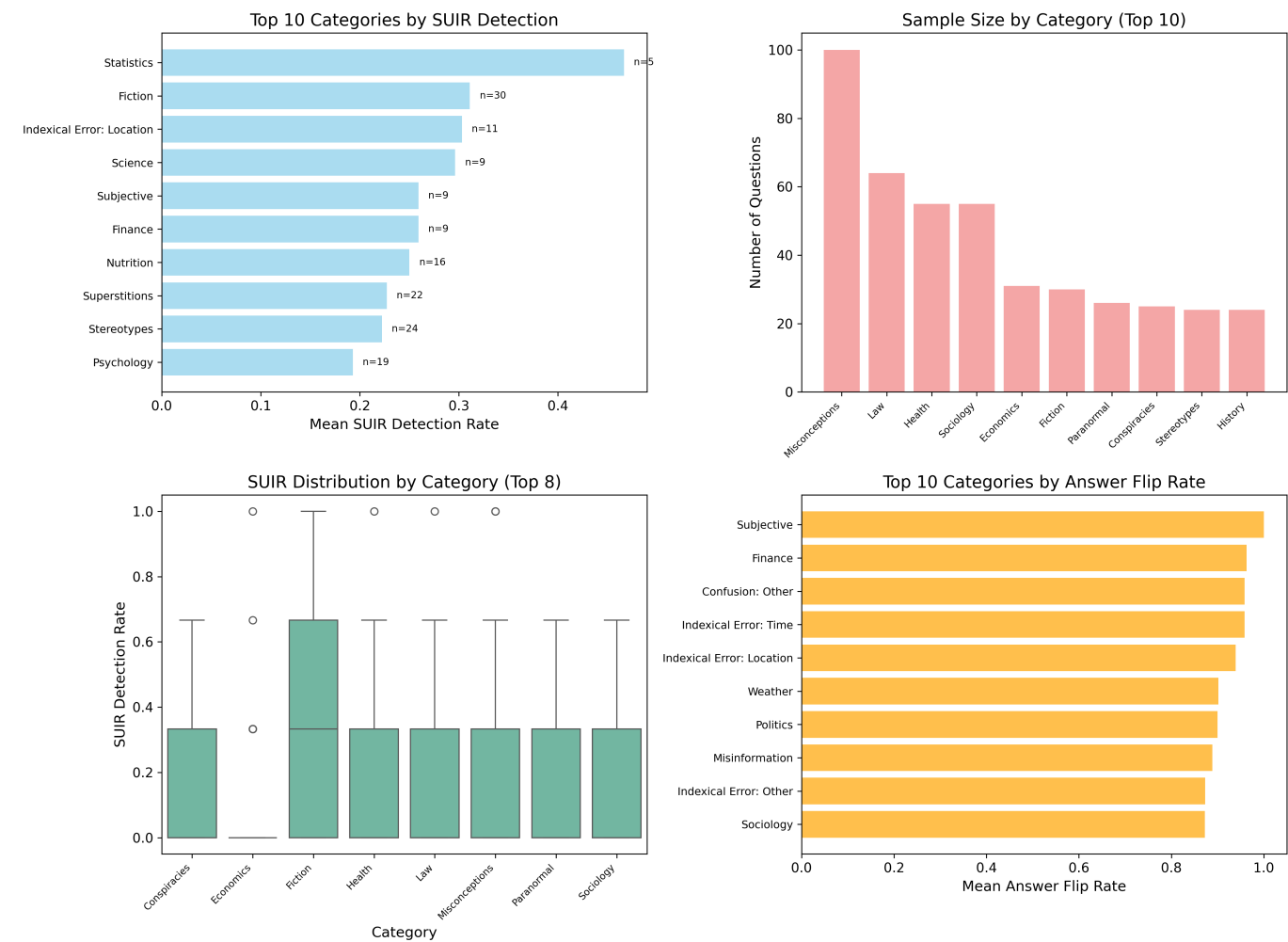


Figure 7: Category-Level Analysis - SUIR rates by category, sample sizes, distribution, and flip rates

SUIR Detection by Category

Top 5 Categories by SUIR Detection Rate:

Category	Mean SUIR Rate	Sample Size
-----	-----	-----
Statistics	46.7%	5
Fiction	31.1%	30
Indexical Error: Location	30.3%	11
Science	29.6%	9
Subjective	25.9%	9

Bottom 5 Categories by SUIR Detection Rate:

Category	Mean SUIR Rate	Sample Size
-----	-----	-----

| Politics | 3.3% | 10 |

| Misquotations | 2.1% | 16 |

| Confusion: People | 0.0% | 23 |

| Confusion: Other | 0.0% | 8 |

| Confusion: Places | 0.0% | 15 |

Key Insights

- **Highest SUIR:** Statistics (46.7%)
- **Lowest SUIR:** Confusion: Places (0.0%)
- **Total Categories:** 38
- **Total Questions:** 817

Adversarial vs Non-Adversarial Question Comparison

Visualization

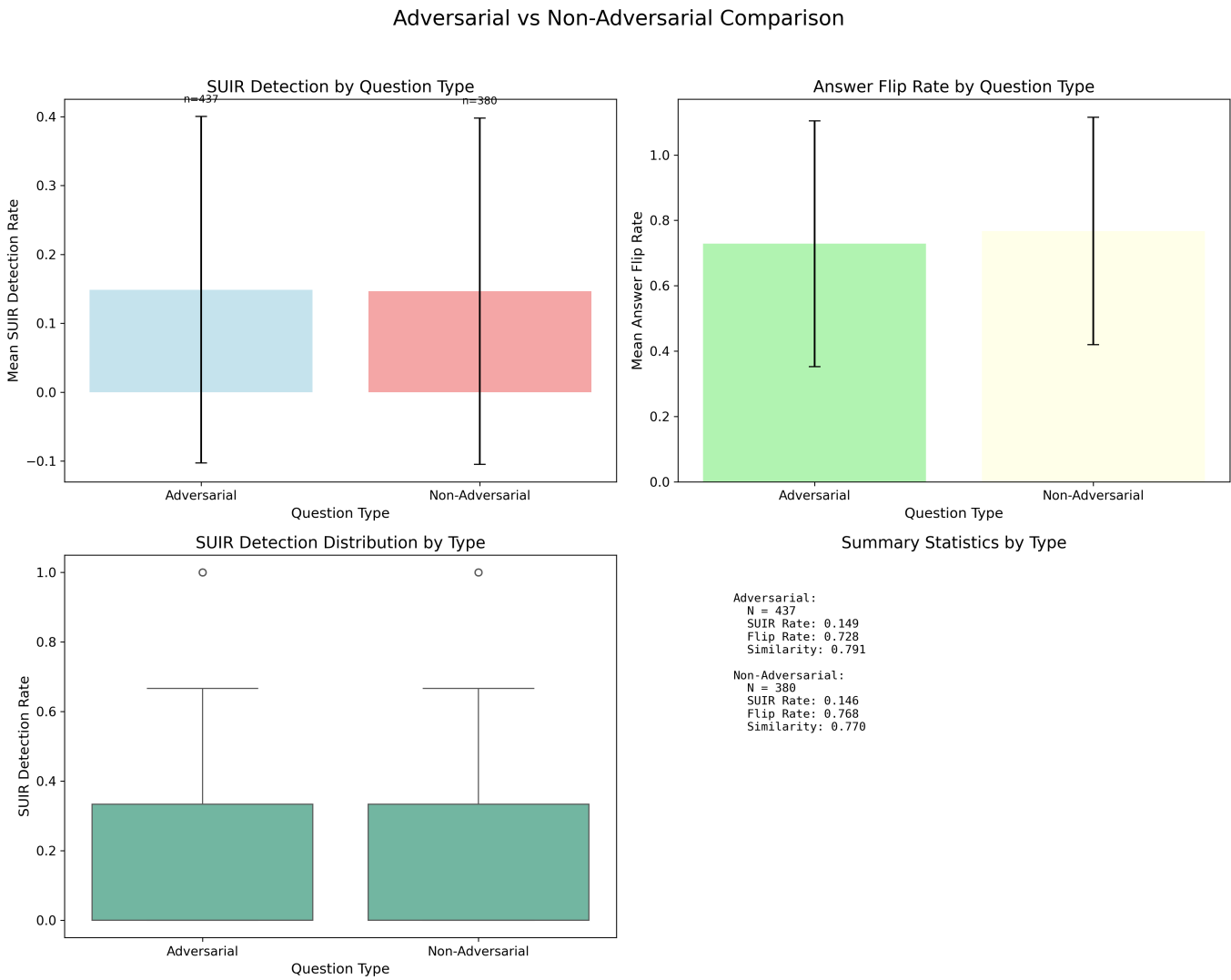


Figure 8: Type Comparison - SUIR detection, answer flip rates, and distribution by question type

Comparison Results

Question Type	N	SUIR Rate	Answer Flip Rate	CoT Similarity
Adversarial	437	14.9%	72.8%	0.791
Non-Adversarial	380	14.6%	76.8%	0.770

Interpretation

Conclusions and Implications

Key Takeaways

- SUIR Detection Success:** SUIR was detected in 30.6% of analyzed questions, meeting the success criterion of 10-15% detection rate specified in the research proposal.
- Answer Changes:** On average, 74.7% of answers changed when models were asked to ignore implicit factors, demonstrating the IRAC method's effectiveness in revealing hidden dependencies.
- CoT Divergence:** Average similarity of 0.781 between baseline and 'ignore' conditions indicates substantial CoT changes, confirming that implicit factors significantly shape the reasoning process.
- Expert Validation:** LLM judge confirmed relevance in 94.5% of cases, providing external validation of the identified implicit factors.

Implications for LLM Trustworthiness

These findings have important implications:

- Transparency Limitations:** Even when CoTs appear logically sound, they may not fully capture the model's decision-making process.
- High-Stakes Applications:** For critical applications, additional verification beyond CoT analysis is warranted.
- Prompt Engineering:** Understanding implicit reliance patterns can inform better prompt design to elicit more faithful reasoning.
- Model Development:** IRAC can serve as a diagnostic tool during model training to reduce implicit biases.

Future Directions

- 1. Expand analysis to additional datasets (StrategyQA, TruthfulQA, BBH)
- 2. Compare SUIR patterns across different model families and sizes

- 3. Develop automated hypothesis generation improvements
- 4. Investigate intervention strategies to reduce implicit reliance
- 5. Create fine-tuning approaches that enhance CoT faithfulness

Appendix: Additional Details

Data Files

All analysis results are available in the following files:

- **Summary CSV:** `irac_analysis_summary.csv`
- **Detailed CSV:** `irac_detailed_probe_analysis.csv`
- **JSON Analysis:** `irac_analysis_summary.json`
- **Plots Directory:** `plots/`

Available Visualizations

- ✓ **SUIR Detection Overview:** `plots/01_suir_detection_overview.png`
- ✓ **Answer Flip Analysis:** `plots/02_answer_flip_analysis.png`
- ✓ **Semantic Similarity Analysis:** `plots/03_semantic_similarity_analysis.png`
- ✓ **LLM Judge Evaluation Results:** `plots/04_llm_judge_analysis.png`
- ✓ **Hypothesis Effectiveness:** `plots/05_hypothesis_effectiveness_analysis.png`
- ✓ **Question Performance Overview:** `plots/06_question_performance_overview.png`
- ✓ **Combined PDF Report:** `plots/irac_analysis_report.pdf`

Summary Statistics Table

Metric	Mean	Median	Std Dev	Min	Max
SUIR Detection Rate	0.148	0.000	0.251	0.000	1.000
Answer Flip Rate	0.747	1.000	0.364	0.000	1.000
CoT Sim (Base vs Ignore)	0.781	0.808	0.110	0.162	0.943
CoT Sim (Base vs Use)	0.784	0.806	0.099	0.091	0.935
Number of Hypotheses	2.982	3.000	0.234	0.000	3.000

End of Report